Patient Journey Intelligence

Introduction, Vision, End-to-End Overview & Value



Tessa Williams
Principal Solution Architect

John Snow LABS

John Snow Labs Overview

A decade-plus of healthcare-specific AI innovation

11th Year

Continued innovation from our expert team, majority with advanced degrees and 2/3 of the company are “do’ers” that use our own models, pipelines, products and solutions.

150M+ Downloads

We provide an open-source version of our core NLP solution that has a huge community and over 150M downloads.

3,000+ Pre-trained Models

Well over 3,000 pre-trained models covering specialty clinical domains in multiple languages.

Expert-led Teams

Data scientists, data engineers, clinicians, researchers, scientists and SMEs driving real-world solutions for healthcare.

100’s of Case Studies and Peer-Reviewed Papers

Our work is validated through hundreds of real-world case studies and peer-reviewed publications.

Award-Winning Innovation

We continue to be recognized for our innovation every year, and we just won our 50th award.

Global Reach & Partnerships

We work with academic research institutions, health systems, life sciences, insurance providers, government and public health, and are integrated with healthcare data platforms, EMR providers, RWD and precision medicine partners.

Proven Across Real-World Use Cases

Early disease detection, at-risk patient identification, treatment pathway and response management, care gap analysis, population health stratification, trends and analytics, disease registries and surveillance, clinical decision support, clinical trial feasibility and prescreening, RWE and comparative effectiveness, precision medicine, rare disease, quality measures and revenue cycle management.

End-to-End Healthcare AI Portfolio

Our solutions started with NLP models and pipelines and have expanded to small language models, our own LLMs, clinical AI model annotation, terminology/ontology services and our latest innovation, Patient Journey Intelligence — a unified platform with agentic and RAG orchestration that leverages healthcare AI to facilitate human experts.

Why Patient Journey Intelligence Exists

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Healthcare has no shortage of data—only a shortage of usable patient intelligence.

EMR
Structured fields

Notes
Clinical narrative

Labs
Results and trends

Medications
Past, current and new therapies

Imaging
DICOM and reports

Pathology
Findings and specimens

Genomics
Variants and biomarkers

Claims
Utilization

External
Registries and literature

Patient-contributed
Wearables and smartphones

Social Determinants
Environment, lifestyle and socioeconomic factors

Standards
International, Country and Private Sources

The patient story is distributed across systems, formats, modalities and time.

Connecting Records Is Not Enough

Changing patient data means changing patient state

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FEB

Pathology
Diagnosis

- Biopsy results and specimen details
- Disease type and stage
- Molecular and histopathology markers

MAR

Genomics
Biomarker

- Genomic variants and biomarkers
- Actionable mutations and precision targets
- Hereditary risk and stratification

APR

Treatment
Initiation

- Therapy selection and regimen
- Start date and line of therapy
- Clinical trial enrollment (if applicable)

JUL

Imaging
Response

- Imaging studies (CT, MRI, PET, etc.)
- Response to treatment (e.g., RECIST)
- Progression, stability or remission

OCT

Progress Note
Progression

- Clinician assessment and patient status
- Clinical progression or improvement
- Next steps in care (treatment change, monitoring, or maintenance)

What happened? → When? → What changed? → What was known? → What happened next?

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Introducing Patient Journey Intelligence

One longitudinal representation—many ways to use it

MULTIMODAL DATA
Bring all patient data together

CLINICAL EXTRACTION
Turn unstructured data into structured, standardized insights

HARMONIZATION
Unify, normalize and enrich across sources

LONGITUDINAL RECORD
Create a complete, time-aware patient view

Understand
Extract meaning from all types of clinical data

Connect
Link findings and events over time

Reason
Healthcare LLMs and grounded tools

Use
Cohorts, registries, research and AI

PJI is the solution layer built upon proven AI products and pipelines
Transforming fragmented data into a trusted, longitudinal patient record to accelerate research, improve care and enable AI.

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Healthcare NLP and Small Language Models

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RAW CLINICAL DATA
Diverse, multimodal and real-world

HEALTHCARE NLP / SLMs
Domain-tuned models and pipelines

EXTRACTED CLINICAL FACTS
Normalized, linked and enriched

NORMALIZED CONCEPTS
Standards-aligned and interoperable

Input
Bring together all types of clinical data

Text, Structured Feeds, PDFs, Images, DICOM

Understand
Extract meaning with domain-specific AI

Entity Recognition, Assertion and Negation, Temporality, Relationships, Entity Linking

Facts
Convert unstructured data into trusted facts

Diagnoses, Medications, Lab Results, Vitals, Procedures, Phenotypes, Genomics and Biomarkers, Outcomes

Standardize
Map to industry standards for interoperability

SNOMED CT, RxNorm, LOINC, ICD, HPO, Additional Ontologies and Terminologies

Data Curation pipelines turn unstructured evidence into reusable clinical meaning

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Raw Clinical Data

RAW CLINICAL DATA
Diverse, multimodal and real-world

Document Types

- **Initial presentation** — H&P, consultation and imaging reports establish symptoms and suspected disease.
- **Diagnosis** — Pathology, cytology and molecular reports establish disease characteristics.
- **Characterization** — Genomics, biomarkers and staging documentation refine the clinical picture.
- **Interventions** — Oncology notes, medication orders, infusion records and procedure reports.
- **Response assessment** — Follow-up imaging, labs, progress notes and treatment assessments document changes.
- **Progression or remission** — Imaging, biomarkers and clinician assessments establish subsequent disease states.
- **Long-term outcomes** — Survivorship notes, follow-up encounters, registries and mortality data support longitudinal outcomes analysis.

Modalities

- **Unstructured Text** – (TXT, CSV)
- **Unstructured Text** – from a DB (CHAR, VARCHAR, NVARCHAR, BLOB)
- **Text from Structured Data** – JSON, XML, HL7, FHIR, CDA, C-CDA
- **Structured Data**
- **PDFs** (machine generated, images)
- **Images**
  DICOM (including slices)

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Healthcare NLP/SLMs

HEALTHCARE NLP / SLMs
Domain-tuned models and pipelines

Input
Multi-Modal Clinical Data

- **Clinical Text** — TXT, PDF, DOC, FHIR
- **Structured Data** — EHR, CSV, JSON
- **Images** — Scanned Documents
- **Imaging** — DICOM & Reports
- **Pathology** — Reports & PDFs

Healthcare NLP / SLMs
Clinical Entity Recognition

- **Entities** — Diagnoses, Medications, Procedures, etc.
- **Linking** — Map to medical terminologies
- **Assertion** — Negation, Uncertainty, Subject, Experiencer
- **Temporality** — When it happened
- **Relationships** — Entity relationships (e.g., treatment for condition)

Model Library
Pre-trained & Custom Models

- **Domain Models** — 3,000+ pre-trained models
- **Specialty Models** — Oncology, Cardiology, Neurology, etc.
- **Custom Models** — Fine-tuned for your use case

Output
Rich Clinical Information

- **Entity Output**
- **Classification (ML)**
- **Coding** — SNOMED, ICD, HPO, CPT, etc.
- **Standardization**
- **Stratification**
- **De-Identification & Pseudo-Anonymization**

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Extracted Clinical Facts

EXTRACTED CLINICAL FACTS

Normalized, linked and enriched

- **Demographics & identity:** Patient identifiers, age, gender, race and ethnicity, encounters, provider and organization relationships
- **Diagnoses & conditions:** ICD-10-CM, SNOMED CT, problem lists, encounter diagnoses
- **Medications:** Prescriptions, active medication lists, administrations (MAR), dispensing, allergies
- **Laboratory observations:** LOINC-coded results, quantitative values, units, reference ranges
- **Vital signs:** Blood pressure, temperature, pulse, oxygen saturation, weight, BMI, etc.
- **Procedures & orders:** CPT/HCPCS, procedure orders, results, surgery, treatments
- **Medical imaging:** DICOM images, series, studies, imaging metadata
- **Pathology & molecular data:** Specimen metadata, structured biomarker results, genomic variants
- **Care plans & workflows:** Appointments, referrals, orders, follow-up tasks, care pathways
- **Devices & monitoring:** Implantable devices, telemetry, wearables, remote patient monitoring
- **Financial & utilization:** Claims, billing codes, length of stay, readmissions, utilization
- **Clinical research:** Trial eligibility, consent, study participation, protocol-related observations

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Normalized Concepts

NORMALIZED CONCEPTS

Standards-aligned and interoperable

- **SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms):** Covers 350,000+ clinical concepts spanning diagnoses, findings, procedures, and anatomical structures.
- **ICD-10-CM / ICD-10-PCS:** The international standard for diagnostic (CM) and procedure (PCS) billing and epidemiology codes.
- **RxNorm:** U.S. National Library of Medicine standardized drug nomenclature linking brand names, generics, and clinical drug ingredients.
- **LOINC (Logical Observation Identifiers Names and Codes):** Universal codes covering 95,000+ laboratory observations, clinical measurements, and diagnostic studies.
- **CPT-4 (Current Procedural Terminology):** Medical services and procedure codes.
- **NDC (National Drug Code):** Standardized product identifiers for medications.
- **HPO (Human Phenotype Ontology):** Phenotypic abnormalities and clinical features used for rare disease and genetics research.
- **UMLS Metathesaurus (Unified Medical Language System):** Cross-terminology mappings that enable translation and semantic alignment across different coding systems.
- **ATC (Anatomical Therapeutic Chemical Classification System):** Drug classification system.
- **MedDRA (Medical Dictionary for Regulatory Activities):** Regulatory activities and adverse event terminology.
- **MeSH (Medical Subject Headings):** Biomedical and health-related terminology indexing system.

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From Extracting to Reasoning

Understand → Reason → Act

UNDERSTAND	REASON	ACT
Healthcare NLP / SLMs	**Healthcare LLMs**	**Agents & Tools**
Extract entities, relations and context	Synthesize, explain and answer questions	Apply approved workflows and tools

Specialized extraction and grounded reasoning are complementary—not competing approaches

Medical Large Language Models

Domain-specific LLMs that combine medical knowledge, clinical reasoning and your data to deliver accurate, relevant and concise answers.

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Chat
Natural language questions

Agents
Task-specific assistants

Agentic Workflows
Automate complex tasks

RAG
Grounded in your data and knowledge

Healthcare LLMs

Domain-Specific • Secure • Your Data

Clinical Knowledge
Guidelines, literature and best practices

Clinical Assessment
Summarize, interpret and explain

Clinical Pathways
Evidence-based recommendations

Anatomy
Anatomical structures and relationships

Life Sciences
Molecular and cellular knowledge

Medical Literature
Search, synthesize and cite sources

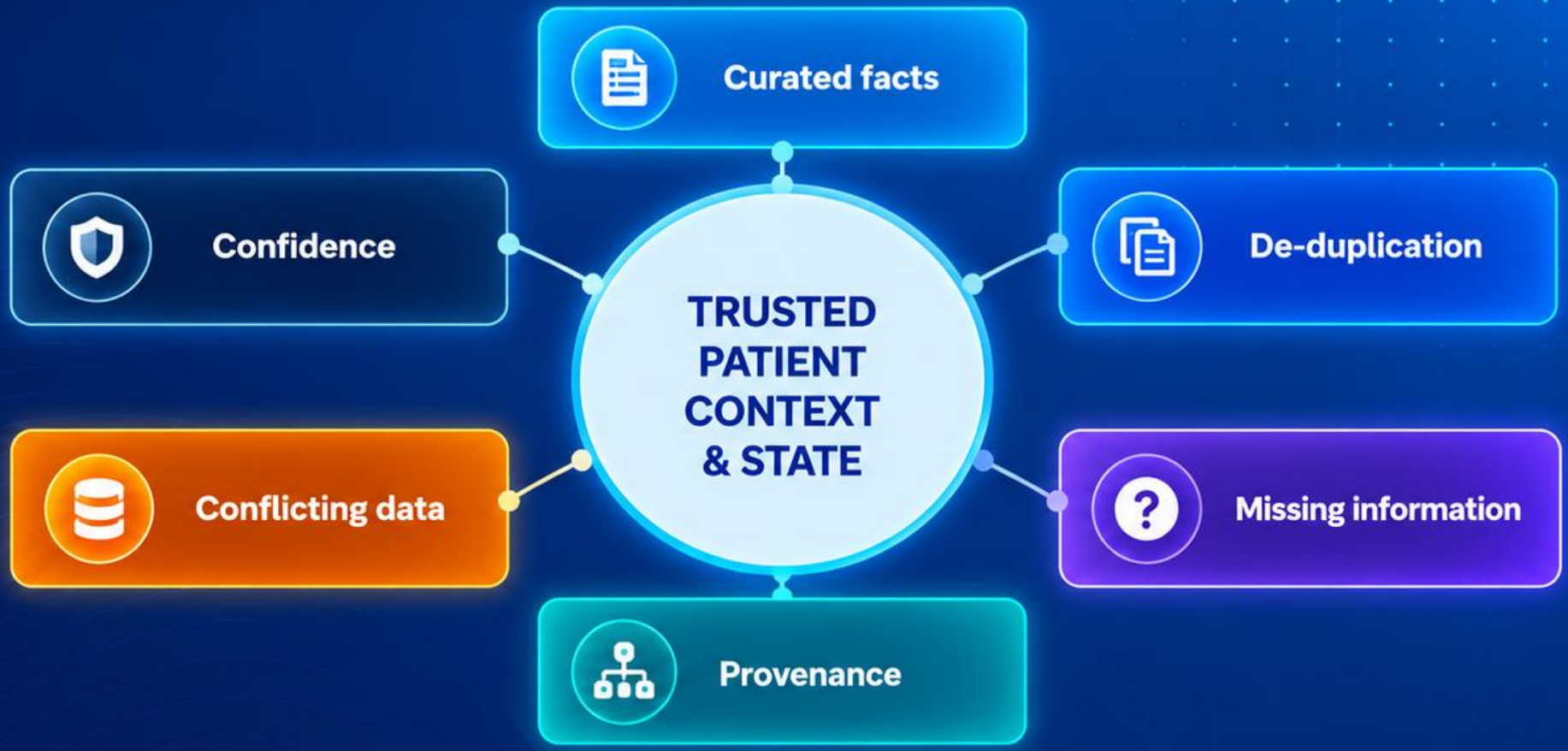
Genetics
Variants, biomarkers and genetic context

Biomedical
Mechanisms, pathways and biological processes

Ontologies & Standards
SNOMED CT, LOINC, ICD, RxNorm and more

An LLM Alone Is Not the Solution

Powerful models still need trustworthy patient context



PJI creates the governed data foundation for natural-language access and agentic AI

PJI Platform Architecture

An integrated platform that turns multi-modal clinical data into a longitudinal, standardized and trusted foundation for research, population health and AI applications.



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Data Curation Studio & Human-in-the-Loop

Let AI facilitate, and put expert attention where it matters

AI EXTRACTS
Clinical entities, assertions and relationships

QUALITY FLAGS
Uncertain, missing or conflicting information

EXPERT REVIEWS
Human-in-the-loop validation and adjudication

APPROVED OUTPUT
High-quality, governed data for downstream use

Evidence
What source supports the assertion?

Exceptions
What is uncertain or conflicting?

Accountability
Who reviewed or changed the result?

AI handles scale. Experts handle judgment and exceptions.

Privacy, Security & Data Access

Private, Secure, Governed

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Secure Deployment & Data Residency

Private deployment, customer-controlled infrastructure, and data residency.

Identity & Access Management

Authentication, role-based permissions, and controlled access to patient data and platform functions.

Data Protection & Privacy

De-identification and protection of sensitive clinical information. Anonymization and pseudonymization.

Governance & Auditability

Data provenance, traceability, review workflows, and audit records.

AI & LLM Security

Private model execution, controlled access to clinical context, and governance of AI-driven workflows.

Secure Data Sharing & Secondary Use

Governed access to curated datasets, patient cohorts, registries, and research outputs.

Security and Privacy is a design and governance decision.

Agents & Tools: Governed AI Applications

Make longitudinal patient intelligence accessible in workflows

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TRUSTED LONGITUDINAL FOUNDATION

Unified, standardized and governed patient data

ASK

Evidence-grounded patient questions

FIND

Cohorts, criteria and patient sets

SUMMARIZE

Timelines and clinical narratives

ORCHESTRATE

Approved tools and tasks

Grounded, governed access to patient intelligence—not unconstrained automation.

Patient Journey Intelligence

The complete patient intelligence lifecycle

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Ingest → Extract → Normalize → Curate → Enrich & Summarize → Leverage

Source modalities

Structured, text, PDF, imaging and DICOM

AI pipeline

Entities, assertions, relations

Standardization

Ontologies, OMOP

Governance & trust

Provenance, HITL, access, de-identification

Patient journey

Events and changes over time – the Patient State

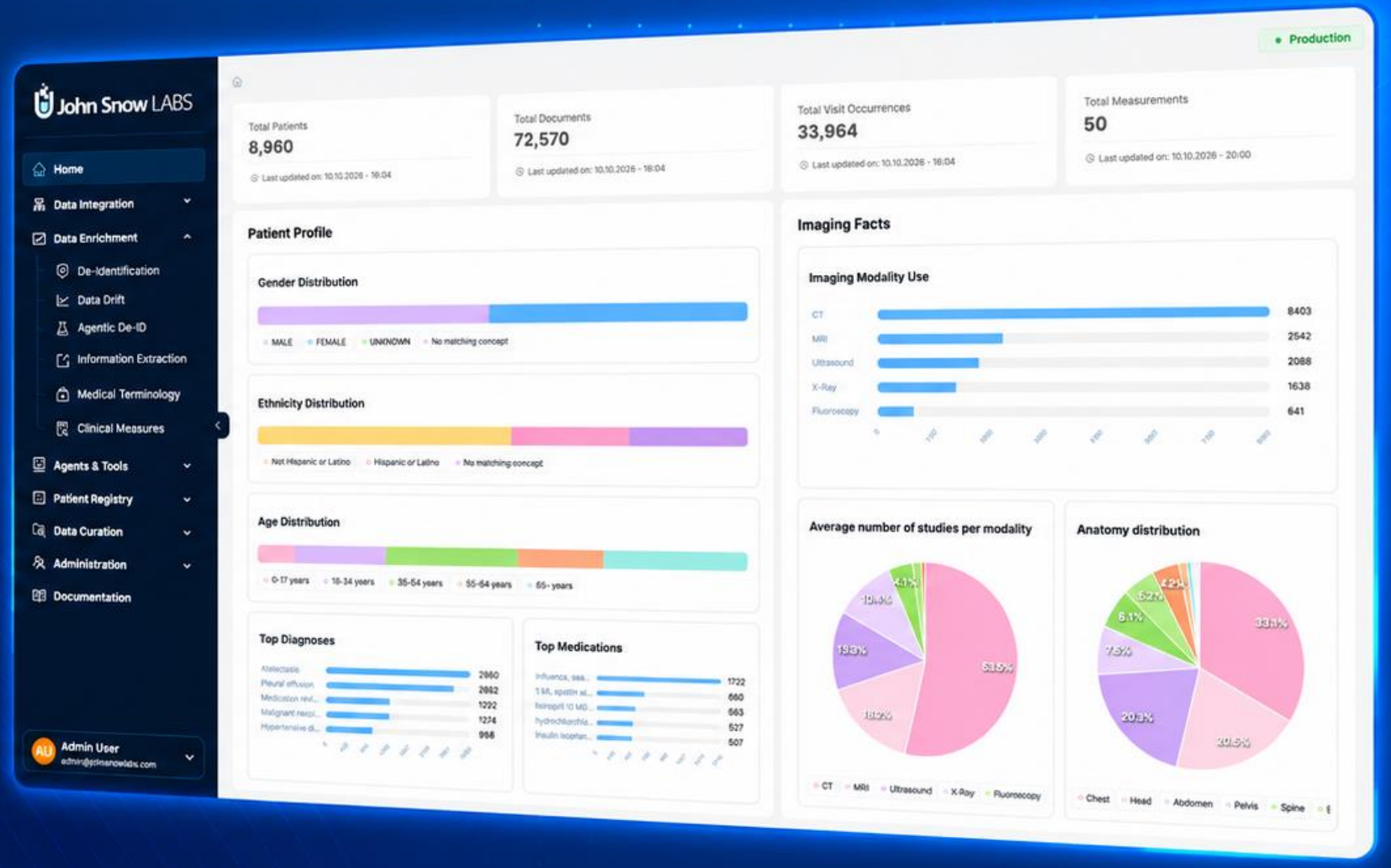
Outcomes

Cohorts, Registries, RWE, Patient Analysis

From Data to Outcomes

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Time for Demo



What Comes Next in the Two-Day Training

You now have the map — next, explore the modules

DATA INTEGRATION

How data enters PJI

DATA ENRICHMENT

How clinical meaning is produced

DATA CURATION STUDIO

How evidence is reviewed and improved

AGENTS & TOOLS

How intelligence is accessed and applied

PATIENT REGISTRY

How structured clinical outputs are used

END-TO-END WORKFLOWS

How modules connect in practice

Today: the map. Next: the mechanics, workflows and applications.

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Thank you for attending our Patient Journey Intelligence Training

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